

1. Overview & specifications

The Frostline Window AC WA-220 (Frostline, air-conditioner) is covered by this service manual for authorised technicians holding the 'general-hvac' skill. It documents diagnosis, fault codes, and board/mechanical-level repair.

Specifications — type: window AC, capacityTon: 1.0, refrigerant: R32, powerW: 1100, starRating: 3.

Warranty: 24 months. Parts and labour on the compressor and PCB. Consumables and cosmetic parts excluded.

Reference: <https://support.example-fieldflow.test/wa-220/manual>

Frostline Window AC WA-220 — Service Manual — page 1

2. Installation & commissioning

Mount the indoor unit level, flare and torque the refrigerant lines, pull a vacuum to 500 microns, then release the R32 charge and confirm subcooling.

Record the serial number and commissioning date against the work order. A failed self-test must be cleared before the visit is closed.

3. Fault codes

H1 — Compressor start capacitor weak; unit hums, no cooling. Replace CAP-220. Skill: general-hvac.

H3 — Control board relay failure. Replace PCB-220. Skill: general-hvac.

H5 — Thermostat stuck; no temperature regulation. Replace THRM-220. Skill: general-hvac.

H9 — Blocked filter; weak airflow (no part needed). No part required; inspect and reset. Skill: general-hvac.

4. Part replacement — PCB-220 / COMP-220

PCB-220 — Window AC control board (board). Isolate mains and follow the safety SOP. Remove the faulty unit, fit PCB-220, and run the self-test. Requires 'general-hvac'. This part is covered under active warranty (free). List price ■2,300.

COMP-220 — Reciprocating compressor (compressor). Isolate mains and follow the safety SOP. Remove the faulty unit, fit COMP-220, and run the self-test. Requires 'general-hvac'. This part is covered under active warranty (free). List price ■9,800.

5. Part replacement — FAN-220 / THRM-220

FAN-220 — Blower fan motor (motor). Isolate mains and follow the safety SOP. Remove the faulty unit, fit FAN-220, and run the self-test. Requires 'general-hvac'. This part is chargeable (out of warranty scope). List price ■1,750.

THRM-220 — Rotary thermostat (thermostat). Isolate mains and follow the safety SOP. Remove the faulty unit, fit THRM-220, and run the self-test. Requires 'general-hvac'. This part is chargeable (out of warranty scope). List price ■680.

6. Part replacement — FILTER-220 / CAP-220

FILTER-220 — Washable mesh filter (consumable). Isolate mains and follow the safety SOP. Remove the faulty unit, fit FILTER-220, and run the self-test. Requires 'general-hvac'. This part is chargeable (out of warranty scope). List price ■320.

CAP-220 — Compressor start capacitor (board). Isolate mains and follow the safety SOP. Remove the faulty unit, fit CAP-220, and run the self-test. Requires 'general-hvac'. This part is covered under active warranty (free). List price ■540.

7. Maintenance, safety & warranty

Maintenance: inspect at each visit; clean or replace consumables on the service interval; log any recurring fault code.

Discharge the high-voltage capacitor before touching the inverter board. Recover refrigerant; never vent R32.

Warranty annex: under an active warranty, PCB-220, COMP-220, CAP-220 are replaced free with labour. Consumables and cosmetic parts are never covered. If warranty is expired or the fault is outside the original scope, the part is chargeable: a quote is raised and payment is collected before the visit is confirmed.

Policy reference: <https://support.example-fieldflow.test/warranty/v5>